

Iterative Method

Definition:

An **iterative method** solves a problem by repeating a set of instructions (loop) until a required condition is satisfied.

Algorithm

1. Start.
2. Initialize the required variables.
3. Repeat the operations inside a loop.
4. Check the condition after each iteration.
5. If the condition is satisfied, stop.
6. Otherwise, continue to the next iteration.
7. End.

Output:

```
Enter a number: 5
Factorial = 120

...Program finished with exit code 0
Press ENTER to exit console.□
```

Time Complexity

The time complexity depends on the number of iterations.

- If the loop runs **n times** → **$O(n)$**
- If there are two nested loops → **$O(n^2)$**

Space Complexity

Iterative methods generally use **$O(1)$** extra space because they do not require additional memory for recursive function calls.

In short:

Iterative = Repetition using loops → usually less memory than recursion.